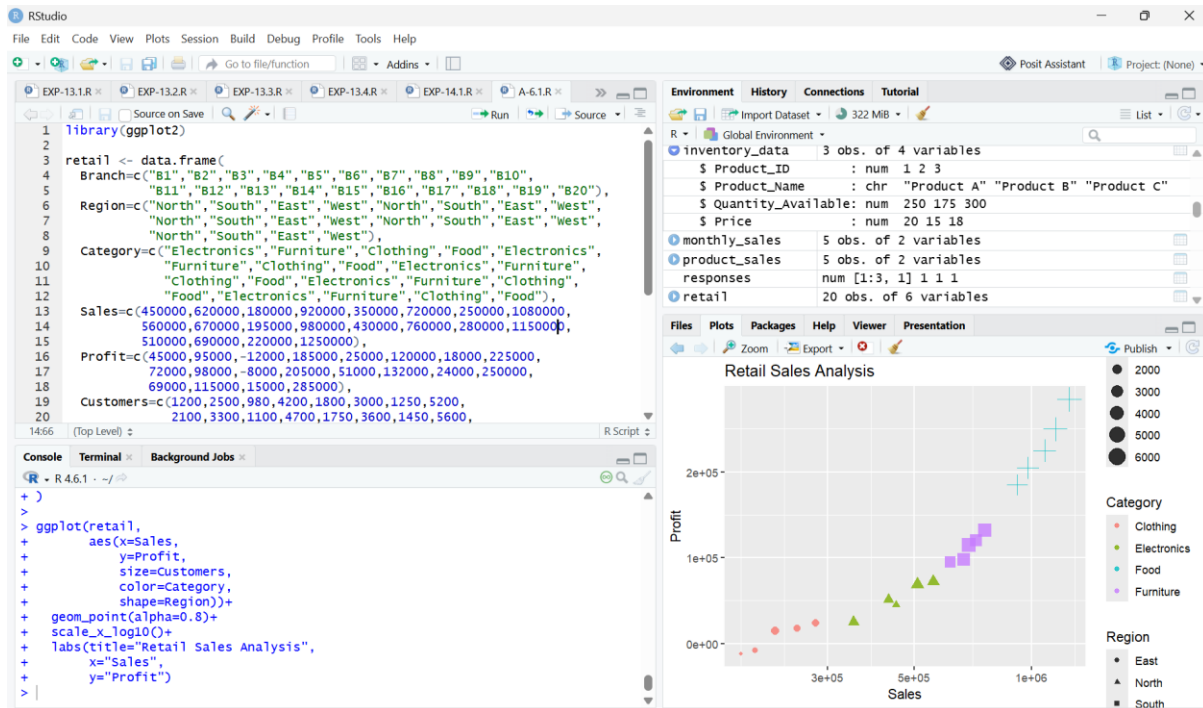


CO1_AT3

QUESTION-1



CODE:

```
library(ggplot2)
```

```
retail <- data.frame(
```

```
  Branch=c("B1","B2","B3","B4","B5","B6","B7","B8","B9","B10",
            "B11","B12","B13","B14","B15","B16","B17","B18","B19","B20"),
  Region=c("North","South","East","West","North","South","East","West",
            "North","South","East","West"),
  Category=c("Electronics","Furniture","Clothing","Food","Electronics",
             "Furniture","Clothing","Food","Electronics","Furniture",
             "Clothing","Food","Electronics","Furniture","Clothing",
             "Food","Electronics","Furniture","Clothing","Food"),
```

```
  Sales=c(450000,620000,180000,920000,350000,720000,250000,1080000,
           560000,670000,195000,980000,430000,760000,280000,1150000,
           510000,690000,220000,1250000),
  Profit=c(45000,95000,-12000,185000,25000,120000,18000,225000,
           72000,98000,-8000,205000,51000,132000,24000,250000,
           69000,115000,15000,285000),
  Customers=c(1200,2500,980,4200,1800,3000,1250,5200,
              2100,3300,1100,4700,1750,3600,1450,5600,
              2100,3300,1100,4700,1750,3600,1450,5600),
)
```

```
  Sales=c(450000,620000,180000,920000,350000,720000,250000,1080000,
           560000,670000,195000,980000,430000,760000,280000,1150000,
```

```

510000,690000,220000,1250000),
Profit=c(45000,95000,-12000,185000,25000,120000,18000,225000,
72000,98000,-8000,205000,51000,132000,24000,250000,
69000,115000,15000,285000),
Customers=c(1200,2500,980,4200,1800,3000,1250,5200,
2100,3300,1100,4700,1750,3600,1450,5600,
2200,3900,1600,6100)
)

```

```

ggplot(retail,
  aes(x=Sales,
    y=Profit,
    size=Customers,
    color=Category,
    shape=Region))+
geom_point(alpha=0.8)+
scale_x_log10()+
labs(title="Retail Sales Analysis",
  x="Sales",
  y="Profit")

```

2. Aesthetic Mapping

Variable	Aesthetic
Sales	X-axis (Log Scale)
Profit	Y-axis
Customers	Bubble Size
Category	Color
Region	Shape

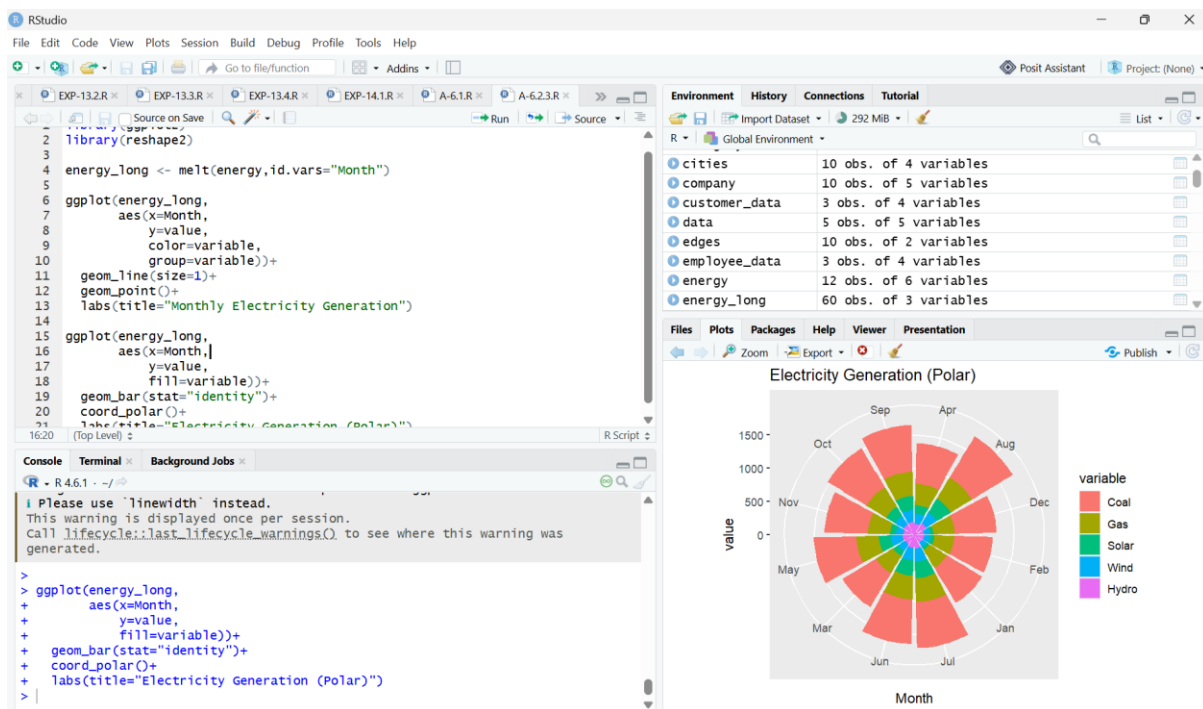
3. Scale Transformation

- **Logarithmic (scale_x_log10())** is used for Sales because the sales values vary widely.
- It spreads data evenly and improves comparison.

4. Justification

- Scatter plot shows relationship between Sales and Profit.
- Bubble size represents customer count.
- Different colors distinguish product categories.
- Shapes separate regions.
- Log scale reduces skewness and improves readability.

QUESTION-2



CODE:

```
library(ggplot2)
```

```
library(reshape2)
```

```
energy_long <- melt(energy, id.vars="Month")
```

```
ggplot(energy_long,  
  aes(x=Month,  
    y=value,  
    color=variable,  
    group=variable))+  
geom_line(size=1)+  
geom_point()+  
labs(title="Monthly Electricity Generation")
```

```
ggplot(energy_long,  
  aes(x=Month,  
    y=value,  
    fill=variable))+  
geom_bar(stat="identity")+  
coord_polar()+  
labs(title="Electricity Generation (Polar)")
```

Comparison

Cartesian

- Easy comparison across months.
- Better for identifying trends.
- More accurate.

Polar

- Attractive visualization.
- Shows seasonal cycle.
- Harder to compare values accurately.

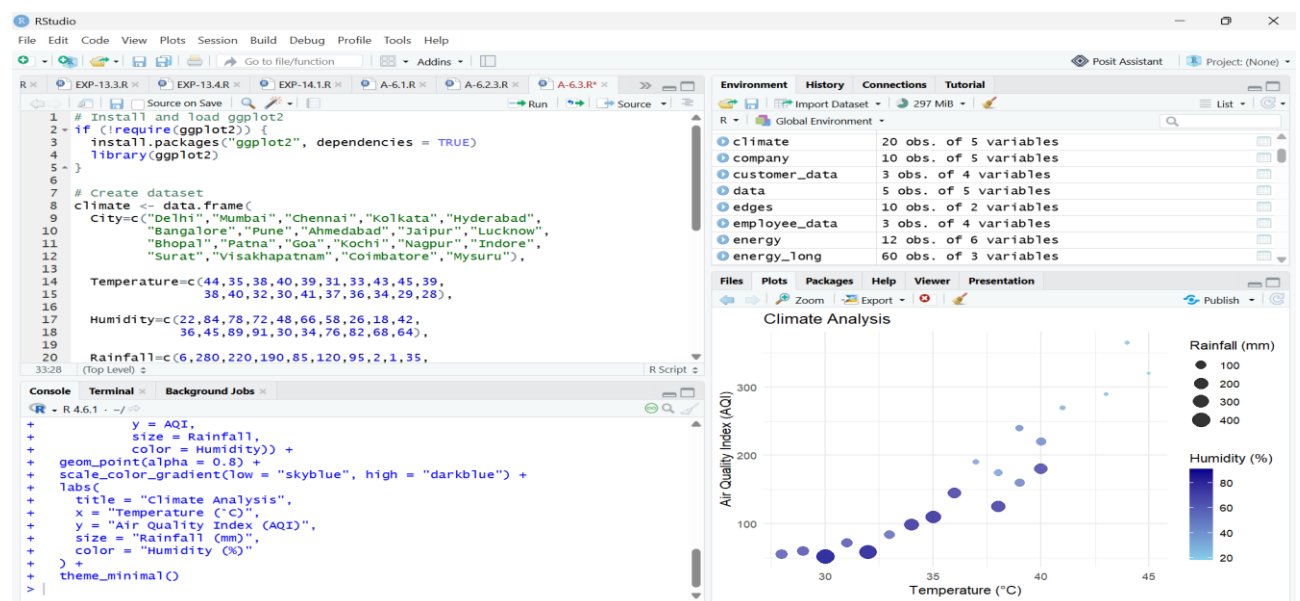
Better Coordinate System

Cartesian Coordinate System

Justification

Electricity generation is a time-series dataset. Cartesian coordinates make trend analysis and month-to-month comparison much easier than polar coordinates.

QUESTION-3



"Surat","Visakhapatnam","Coimbatore","Mysuru"),

Temperature=c(44,35,38,40,39,31,33,43,45,39,
38,40,32,30,41,37,36,34,29,28),

Humidity=c(22,84,78,72,48,66,58,26,18,42,
36,45,89,91,30,34,76,82,68,64),

Rainfall=c(6,280,220,190,85,120,95,2,1,35,
48,72,360,410,12,20,180,240,130,145),

AQI=c(365,110,125,180,160,72,84,290,320,240,
175,220,58,52,270,190,145,98,60,55)

)

Plot

```
ggplot(climate,  
  aes(x = Temperature,  
      y = AQI,  
      size = Rainfall,  
      color = Humidity)) +  
  geom_point(alpha = 0.8) +  
  scale_color_gradient(low = "skyblue", high = "darkblue") +  
  labs(  
    title = "Climate Analysis",  
    x = "Temperature (°C)",  
    y = "Air Quality Index (AQI)",  
    size = "Rainfall (mm)",  
    color = "Humidity (%)"
```

) +

theme_minimal()

Aesthetic Mapping

Variable	Aesthetic
-----------------	------------------

Temperature	X-axis
-------------	--------

AQI	Y-axis
-----	--------

Rainfall	Bubble Size
----------	-------------

Humidity	Color (Viridis)
----------	-----------------

Colour Palette

Viridis

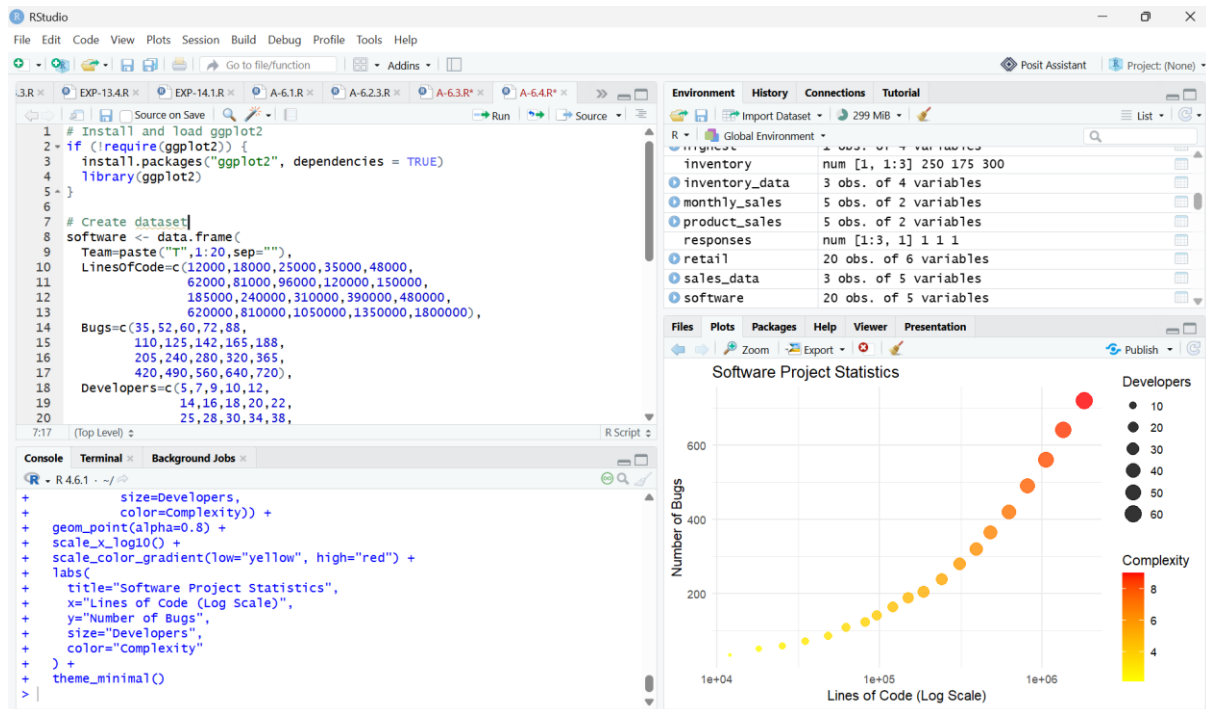
Reason:

- Color-blind friendly
- Perceptually uniform
- Easy to interpret
- Good for continuous variables

Justification

- Temperature and AQI relationship becomes clear.
- Bubble size shows rainfall intensity.
- Color gradient represents humidity effectively.

QUESTION-4




```
205,240,280,320,365,  
420,490,560,640,720),  
Developers=c(5,7,9,10,12,  
14,16,18,20,22,  
25,28,30,34,38,  
42,46,50,55,60),  
Complexity=c(2.1,2.4,2.7,3.0,3.2,  
3.5,3.8,4.1,4.5,4.8,  
5.1,5.4,5.8,6.2,6.6,  
7.1,7.6,8.0,8.5,9.0)  
)
```

```
# Visualization
```

```
ggplot(software,  
  aes(x=LinesOfCode,  
    y=Bugs,  
    size=Developers,  
    color=Complexity)) +  
  geom_point(alpha=0.8) +  
  scale_x_log10() +  
  scale_color_gradient(low="yellow", high="red") +  
  labs(  
    title="Software Project Statistics",  
    x="Lines of Code (Log Scale)",  
    y="Number of Bugs",  
    size="Developers",  
    color="Complexity"  
  ) +  
  theme_minimal()
```

Selected Transformation

Logarithmic Transformation (scale_x_log10())

Justification

The Lines of Code values range from **12,000 to 1,800,000**, making the data highly skewed.

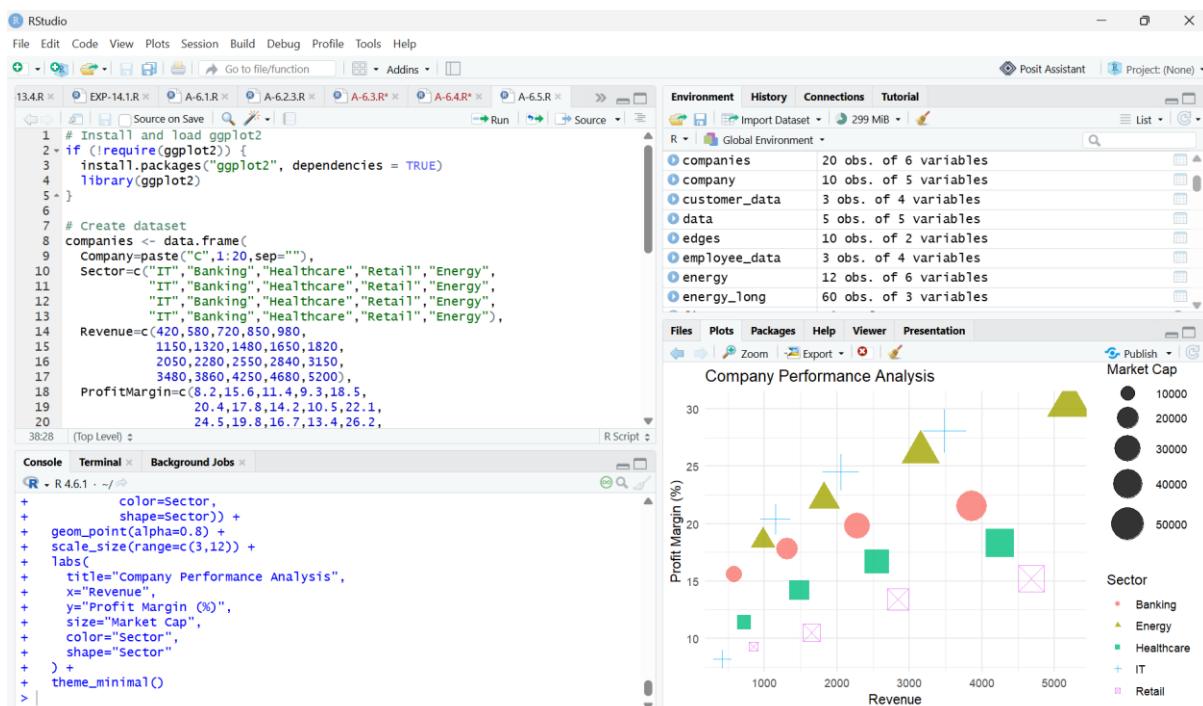
A logarithmic scale:

- Compresses large values.
- Improves visibility of smaller projects.
- Reveals hidden trends.
- Produces better spacing than a linear scale.

Why Better than Linear Scale?

- Prevents data clustering.
- Makes comparison easier.
- Reduces skewness.
- Improves readability.

QUESTION-5



CODE:

```
# Install and load ggplot2

if (!require(ggplot2)) {
  install.packages("ggplot2", dependencies = TRUE)
  library(ggplot2)
}

# Create dataset

companies <- data.frame(
  Company=paste("C",1:20,sep=""),
  Sector=c("IT","Banking","Healthcare","Retail","Energy",
"IT","Banking","Healthcare","Retail","Energy",
"IT","Banking","Healthcare","Retail","Energy",
"IT","Banking","Healthcare","Retail","Energy"),
  Revenue=c(420,580,720,850,980,
1150,1320,1480,1650,1820,
2050,2280,2550,2840,3150,
3480,3860,4250,4680,5200),
  ProfitMargin=c(8.2,15.6,11.4,9.3,18.5,
20.4,17.8,14.2,10.5,22.1,
24.5,19.8,16.7,13.4,26.2,
28.1,21.5,18.3,15.2,30.4),
  Employees=c(1500,3200,2100,4200,2800,
5100,6800,3900,7200,4500,
8100,9500,6200,10300,7200,
11800,13200,8600,14500,16000),
  MarketCap=c(8500,12000,9800,7600,14500,
18200,21500,16800,13200,24500,
28500,32500,24800,19200,38200,
```

```
42500,46800,35600,29800,52000)
```

```
)
```

```
# Visualization
```

```
ggplot(companies,
```

```
  aes(x=Revenue,
```

```
      y=ProfitMargin,
```

```
      size=MarketCap,
```

```
      color=Sector,
```

```
      shape=Sector)) +
```

```
geom_point(alpha=0.8) +
```

```
scale_size(range=c(3,12)) +
```

```
labs(
```

```
  title="Company Performance Analysis",
```

```
  x="Revenue",
```

```
  y="Profit Margin (%)",
```

```
  size="Market Cap",
```

```
  color="Sector",
```

```
  shape="Sector"
```

```
) +
```

```
theme_minimal()
```

Aesthetic Mapping

Variable	Aesthetic
Revenue	X-axis
Profit Margin	Y-axis
Market Capitalization	Bubble Size

Variable	Aesthetic
Industry Sector	Color & Shape
Employee Strength	<i>(Optional improvement: add labels using <code>geom_text(aes(label=Employees), vjust=-0.5)</code> or use faceting if needed.)</i>

Coordinate System

Cartesian Coordinate System

Reason:

- Best for comparing continuous variables.
- Clearly shows relationships between revenue and profit margin.

Colour Palette

Use **Viridis** or **Set2**.

- Viridis → Continuous variables
- Set2 → Categorical sectors

Justification

- Revenue vs Profit Margin reveals company performance.
- Bubble size highlights Market Capitalization.
- Colors and shapes distinguish industry sectors.